

CDE2605R Undergraduate Research Experience (UREx)

Smart Maintenance System for Floating Solar Panels: Vision-Guided Selective Cleaning via Gravity-Assisted Mechanism

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1 Abstract

Floating photovoltaic (FPV) systems represent a growing segment of solar energy deployment, particularly in land-constrained, high-irradiance regions such as coastal Singapore. Their performance in coastal or high-humidity environments is significantly degraded by soiling — most critically salt crystallisation — which reduces panel light transmittance and electrical output over time. Existing cleaning solutions, both manual and automated, are energy-intensive, mechanically inflexible, or operate on fixed schedules without sensing actual contamination state.

This report presents an integrated smart maintenance system designed and built by a four-member undergraduate research team over two academic semesters. The system combines a Raspberry Pi camera-based image acquisition pipeline, a YOLOv8 deep learning object detection model, a normalised-coordinate mapping module, and a custom-fabricated gravity-assisted mechanical cleaning mechanism comprising a single top-edge rail for lateral positioning and a spool-cable deployment for vertical reach. An end-to-end orchestration script coordinates image capture on the Pi, Wi-Fi transmission to a laptop for detection and coordinate computation, and autonomous motor actuation to drive a rotating sponge to detected contamination sites.

Experimental validation yielded a detection precision of approximately **0.82** and recall of **0.78** across **40** test images, with coordinate mapping achieving a mean 2D error of approximately **6.5 mm** across **10** reference points. A total of **18** cleaning trials were conducted spanning light, moderate, and heavy contamination levels, achieving a mean removal rate of approximately **78%**.

Limitations including glare-induced false positives and single-panel prototype scope are documented with a concrete development roadmap proposed for future iterations.

2 Acknowledgements

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4. Introduction

4.1 Background and Motivation

Solar photovoltaic (PV) systems have become one of the fastest-growing energy technologies globally, driven by falling costs and increasing environmental awareness. Among PV deployment configurations, floating photovoltaic (FPV) systems are attracting significant interest for their ability to address land scarcity while simultaneously benefiting from water-surface cooling that can raise energy yield by 5–15% compared to equivalent ground-mounted installations.

However, the performance of PV systems is sensitive to a broad range of environmental factors. Fouad et al. [1] provide a comprehensive classification of these factors, identifying solar irradiance, module temperature, dust accumulation, shading, and soiling as the key environmental concerns with major effects on system output. In humid and coastal environments, soiling — particularly salt crystallisation — presents an especially acute maintenance challenge. Alshareef [2] documents that some locations experience power losses of over 1% per day and up to 80% monthly efficiency reduction due to soiling, making regular maintenance essential to preserving economic viability.

4.2 Problem Definition

Existing cleaning approaches fail to address the specific demands of FPV salt crystallisation maintenance. Four specific gaps motivate this project:

- **Lack of selectivity:** Conventional robotic cleaners traverse the full panel surface regardless of contamination distribution. Salt crystallisation is spatially heterogeneous, making full-surface traversal energetically wasteful.

- **Mechanical inflexibility:** Conventional rail-based cleaning robots require precision-aligned horizontal rails spanning the full length of both panel edges, making installation panel-configuration-specific and poorly suited to floating arrays where access is constrained and installation complexity must be minimised. The system proposed in this project reduces this to a single short rail mounted only along the top edge of the panel, with vertical positioning handled by the gravity-assisted spool mechanism — significantly reducing mechanical footprint and installation complexity compared to existing approaches.

- **Absence of intelligence:** Current systems operate on fixed time schedules without sensing actual contamination state. Vision-guided, demand-driven operation is needed to match cleaning effort to real soiling conditions.

- **Lack of salt-specific detection models for floating PV:** Existing computer vision models for PV panel inspection have predominantly been trained to detect general dust accumulation or bird droppings. Salt crystallisation presents a visually distinct contamination type — white crystalline patches with high specular reflectance — that is not adequately represented in most published datasets. At the time this project commenced in August 2025, no publicly available detection model specifically targeting salt residue on floating PV panels existed, necessitating the construction of a custom dataset and trained model. Concurrent work by Li et al. [11], published during the course of this project, has since addressed salt deposition detection on offshore photovoltaic modules using an enhanced YOLOv8 framework — confirming the technical direction pursued here while differing in deployment scope, targeting

fixed offshore installations subject to seawater splash rather than the floating coastal FPV context addressed in this project.

4.3 Project Objectives

This project aims to design, fabricate, and validate an integrated smart maintenance system with five specific objectives:

- Develop a computer vision detection pipeline capable of localising salt crystallisation on FPV panel surfaces with precision sufficient to guide physical cleaning.
- Design and fabricate a gravity-assisted mechanical cleaning mechanism requiring only a single top-edge rail, avoiding the full panel-length dual-rail configurations of conventional robotic cleaning systems.
- Implement a coordinate mapping module translating image-space detections to physical actuation commands.
- Evaluate end-to-end system performance through controlled experiments, quantifying detection accuracy, coordinate mapping error, cleaning effectiveness, and energy consumption.
- Identify limitations and propose a concrete development roadmap for future iterations.

4.4 Scope

This project focuses on salt crystallisation as the primary target contaminant. Hardware prototyping and testing are conducted in a laboratory environment using PV panel mockups; open-water deployment is outside current scope. The cleaning mechanism is evaluated on a single-panel prototype, with multi-panel scalability discussed in the design rationale and future work sections.

5. Literature Review

5.1 Impact of Soiling on PV Performance

The degradation of photovoltaic output due to surface contamination is a well-established phenomenon, yet its severity is frequently underestimated in system-level energy models. Fouad et al. [1] provide a systematic classification of all factors influencing PV performance, positioning soiling alongside irradiance and temperature as a primary operational variable rather than a secondary maintenance concern. This framing is important: it places cleaning not as an optional intervention but as an integral part of sustained energy yield. Alshareef [2] substantiates this with field data, reporting transmission losses of up to 64% at low tilt angles after 38 days of uninterrupted exposure, with daily power losses exceeding 1% in heavily polluted environments. Said et al. [6] corroborate these findings and further demonstrate that the degradation mechanism is primarily optical — soiling reduces light transmittance through the cover glass — with the effect compounded by high ambient humidity and temperature, conditions that are persistently present in coastal Singapore and directly relevant to floating PV deployments in the region.

What distinguishes salt crystallisation from other soiling types is not merely its optical effect but its physical adhesion mechanism. Unlike loose particulate dust, which may be partially removed by wind or rainfall, salt deposits form crystalline structures that bond chemically to the glass surface and require mechanical agitation to dislodge. Shaik et al. [7] experimentally quantify the performance impact of adhesive surface contaminants — including soil deposits and bird droppings — demonstrating measurable degradation in both thermal efficiency and electrical output across a range of tilt angles from 0° to 12.91°, with tilt angle found to have only minor influence on overall contamination accumulation. This last finding has a direct implication for FPV systems: low-tilt floating installations cannot rely on gravity-assisted self-cleaning as a mitigation strategy. Bdour et al. [8], whose primary focus is microcrack-induced power degradation, note in passing that regular cleaning every one to two weeks is recommended to avoid efficiency losses from soiling — a frequency that renders manual intervention economically impractical at scale and argues strongly for automated solutions.

5.2 Conventional Cleaning Technologies

The existing literature on automated PV cleaning reveals a field that has made meaningful progress in physical cleaning mechanisms while remaining almost entirely blind to the question of when and where cleaning is needed. Nkinyam et al. [9] represent the current state of the art in waterless automated cleaning: their Arduino-controlled system combining an air ejector blower with a flexible rolling brush achieved a 26.23% mean power output improvement on a 60 W panel, validating the effectiveness of mechanical contact cleaning without water infrastructure. Constantin et al. [10] further argue that preventive maintenance strategies — particularly those informed by sensor or imaging data — outperform reactive or time-scheduled approaches in terms of energy yield and maintenance cost. Yet despite this argument, the automated systems described in the literature uniformly operate on fixed time intervals, cleaning the entire panel surface on a pre-set schedule regardless of actual contamination state. This represents a fundamental inefficiency: energy and mechanical wear are consumed cleaning areas that are already clean, while genuinely soiled regions receive no more attention than pristine ones.

The mechanical designs of existing automated systems also exhibit a recurring limitation: they are architecturally rigid. Rail-based cleaning robots, the predominant design paradigm, require precision-aligned rails spanning the full perimeter of each panel. On land-based installations with fixed, flat array geometries this is manageable, but on fixed floating arrays — where panels are mounted on anchored pontoon structures over water, making installation and maintenance access more constrained than terrestrial systems — full-perimeter rail alignment represents a significant practical burden. Said et al. [6] note that adaptability and low installation complexity are critical requirements for cost-effective FPV maintenance, yet this requirement goes unaddressed in the mechanical designs reviewed. The system proposed in this work responds to this limitation directly, reducing the rail infrastructure to a single top-edge lateral track and using a gravity-assisted spool-cable mechanism for vertical reach.

5.3 Computer Vision for PV Inspection

Computer vision for PV inspection has developed along two largely separate tracks: thermal infrared imaging for electrical fault detection, and RGB imaging for surface soiling assessment. Di Tommaso et al. [12] exemplify the infrared approach, applying a multi-stage YOLOv3-based pipeline to UAV-captured thermal and visible imagery across two large Italian PV plants, achieving AP@0.5

exceeding 98% for panel detection and a mAP@0.5 of almost 70% for visible-spectrum anomalies including soiling and bird droppings. Critically, their work establishes that this level of detection accuracy is operationally sufficient to guide PV maintenance decisions — a benchmark that carries direct relevance to this project's performance targets. The choice of visible-spectrum RGB imaging over thermal infrared in this project is deliberate: thermal cameras remain prohibitively expensive for embedded deployment on a low-cost prototype, while RGB sensors are readily available and, as Cavieres et al. [13] demonstrate, contain sufficient spectral information to discriminate soiling from clean panel substrate with CNN-based analysis.

Within the RGB soiling detection literature, the dominant contaminants studied have historically been dust and bird droppings. The most directly comparable recent work is that of Li et al. [11], who present an enhanced YOLOv8 framework specifically for salt deposit detection on offshore photovoltaic modules. Their architecture integrates the SimAM parameter-free attention mechanism at the end of the backbone and within the neck feature-fusion layers, and replaces the default CIoU regression loss with WIoU to mitigate harmful gradients introduced by low-quality samples. On a purpose-built dataset of 500 images — constructed by combining real field captures from an offshore floating photovoltaic installation with laboratory-simulated NaCl crystallisation samples — their model achieves a mAP50 of 85.8%, representing a 3% improvement over baseline YOLOv8 while maintaining an inference speed of 67 frames per second. Their dataset construction methodology, combining field imagery with controlled laboratory salt crystallisation, closely parallels the approach adopted in this project and independently validates YOLOv8 as an appropriate backbone for salt-specific contamination detection. Pathak et al. [14] further demonstrate that deep learning approaches can localise hotspots and classify fault types, eliminating the need for manual monitoring at scale.

Despite this recent progress, a contextual gap remains. Salt deposits are visually distinct from dust and bird droppings: they appear as white crystalline patches with pronounced specular reflectance, and their texture and boundary characteristics differ fundamentally from the brown or grey tonal signatures that dominate the majority of published training datasets. Models trained on dust and bird dropping datasets therefore generalise poorly to salt crystallisation. Although Li et al. [11] have since addressed salt-specific detection in the offshore context, the floating photovoltaic deployment targeted in this project presents distinct visual conditions — variable water-surface reflectance, lower tilt angles typical of Singaporean FPV installations, and persistently high coastal humidity — that warrant a domain-specific training dataset and model tailored to these conditions.

5.4 Research Gap

The foregoing review reveals a structural disconnect in the literature that has persisted despite progress in both sub-fields. Computer vision researchers have produced increasingly capable soiling detection systems, but these systems stop at detection: they produce reports or annotations and leave remediation to human operators. Automated cleaning engineers, conversely, have produced increasingly capable physical cleaning mechanisms, but these mechanisms operate blind — executing fixed-schedule full-surface traversals with no knowledge of where contamination actually exists. The two bodies of work have advanced in parallel without converging.

This disconnect has a second, more specific dimension in the context of floating solar in coastal environments. Until the concurrent work of Li et al. [11], published during the course of this project,

no detection model had specifically targeted salt crystallisation as a distinct contamination class — and even now, no published model is trained on floating-panel visual conditions as distinct from fixed offshore installations. The consequence is that a detection–cleaning integration attempted using the broader published model base would be miscalibrated for the primary contaminant most likely to be encountered in floating coastal installations. This project addresses both dimensions simultaneously — building a closed-loop system that connects vision-guided detection to physical actuation, and doing so with a detection model purpose-trained on salt crystallisation in a floating panel context rather than adapted from models built for general soiling or fixed offshore installations. Together, these contributions position this work at the intersection of two previously disjoint research streams and respond directly to the operational realities of FPV maintenance in high-humidity coastal environments.

6. System Design and Methodology

6.1 System Architecture

The proposed system integrates three interdependent subsystems in a closed-loop inspection–detection–cleaning cycle:

- **Vision Subsystem:** Raspberry Pi camera module captures the panel surface as a series of equal-sized image tiles, each covering a defined physical sub-region of the panel. Since the full panel is too large to be captured at sufficient resolution in a single shot, the panel is divided into tiles and each tile is captured individually. The captured images are transmitted to the laptop via TCP over the local Wi-Fi network.

- **Intelligence Subsystem:** A YOLOv8-based detection pipeline running on the laptop preprocesses the received images, identifies contamination regions, and maps detected bounding box centroids to physical panel coordinates in centimetres.

- **Mechanical Subsystem:** A stepper motor-driven spool deploys the cleaning module to target coordinates; the cleaning module removes deposits using a high-speed rotating sponge.

The three subsystems are coordinated by fpv_main_orchestrator.py — a single Python script with two roles ("pi" and "laptop") that orchestrates the full scan–detect–compute–actuate cycle. Figure 1 below shows the data and control flow between subsystems.

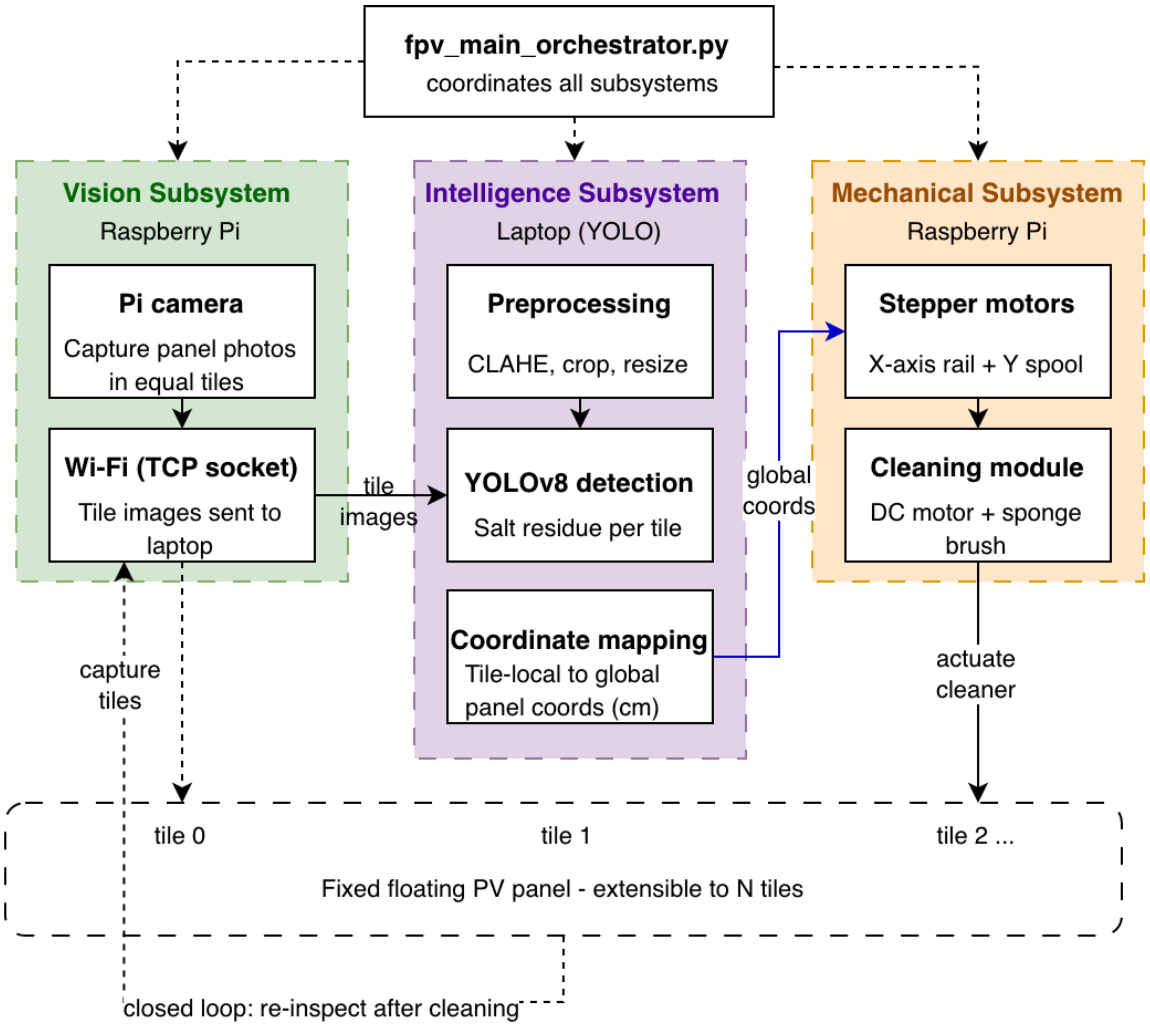


Figure 1: System architecture and data flow

6.2 Vision and Detection Pipeline

6.2.1 Camera Setup and Multi-Image Strategy

The Raspberry Pi Camera Module v2 is mounted on the rail-based cleaning mechanism at a fixed standoff distance from the panel surface. The solar panel is installed at an inclination of approximately 10°, while the camera is positioned at an oblique angle of approximately 47° relative to the panel surface. The camera translates along the rail and captures images at predefined positions corresponding to each panel tile.

This configuration establishes a consistent geometric relationship between the camera and the panel across inspection cycles, which is essential for reliable perspective transformation, coordinate mapping, and repeatable inspection. The oblique imaging geometry inherently introduces perspective distortion, which is subsequently corrected in the preprocessing stage.

Because the full panel surface cannot be captured at sufficient resolution in a single image, a tiling strategy is adopted. The panel is divided into equal-sized tiles, and the camera captures each tile sequentially, with each tile corresponding to a known physical sub-region of the panel. In the current implementation, the panel is divided into four horizontal tiles (tile 0 to tile 3), resulting in a group of images per inspection cycle.

This multi-image strategy enables complete panel coverage while maintaining a close capture distance, allowing small contamination features such as salt crystal deposits to be resolved. The use of fixed capture positions further ensures consistent field-of-view geometry, simplifying downstream geometric transformation and coordinate mapping.

Images are captured using rpicam-still at a resolution of 3840×2464 pixels and transmitted to a laptop over a TCP connection on a local Wi-Fi network using a framed binary protocol consisting of an 8-byte length header followed by payload chunks. Under local hotspot conditions, the average transmission latency per image pair is approximately 8–10 seconds.

The image acquisition stage is executed prior to any cleaning operation, enabling the system to perform inspection, detection, and decision-making in a sequential and integrated workflow.

6.2.2 Preprocessing Pipeline

To improve the robustness and accuracy of contamination detection, a structured preprocessing pipeline is applied to each tile image prior to object detection. The pipeline is implemented using OpenCV and is designed to address common challenges in outdoor photovoltaic imaging, including illumination variability, colour imbalance, and specular surface reflections.

First, a gray-world white balance algorithm is applied to correct colour bias. This method assumes that the average colour of a scene should be neutral and adjusts the red, green, and blue channels accordingly, ensuring that contamination features are not distorted by sensor bias or variations in lighting conditions.

Next, exposure normalisation is performed to reduce brightness variation across images captured under different environmental conditions. A global gain factor is computed based on the mean grayscale intensity and applied to the image to achieve a predefined target brightness. This ensures consistent illumination conditions for the detection model.

A glare reduction step is then applied to address strong specular reflections from the panel glass surface. Regions with high brightness and low saturation in the HSV colour space are identified as glare regions, and their intensity is selectively attenuated using a Gaussian-based smoothing approach. This step mitigates a key failure mode in RGB-based inspection, where reflective regions may otherwise be misclassified as contamination.

Following glare suppression, contrast enhancement is performed using Contrast Limited Adaptive Histogram Equalisation (CLAHE) on the luminance channel. This enhances local contrast while preventing excessive noise amplification, allowing subtle contamination features to be more clearly distinguished.

Finally, geometric correction is performed through a perspective transformation. Due to the inclined placement of the solar panel, captured images exhibit perspective distortion. Four predefined

reference points corresponding to the panel corners are used to compute a homography matrix, mapping the image to a normalised rectangular plane with a fixed output resolution of 1600×900 pixels. Separate sets of reference points are used for tile 0 and tile 1 to account for differences in viewing geometry at each capture position.

As shown in Figure 2, the output of this preprocessing pipeline is a geometrically rectified and photometrically normalised image, providing a consistent and reliable input for the detection model.

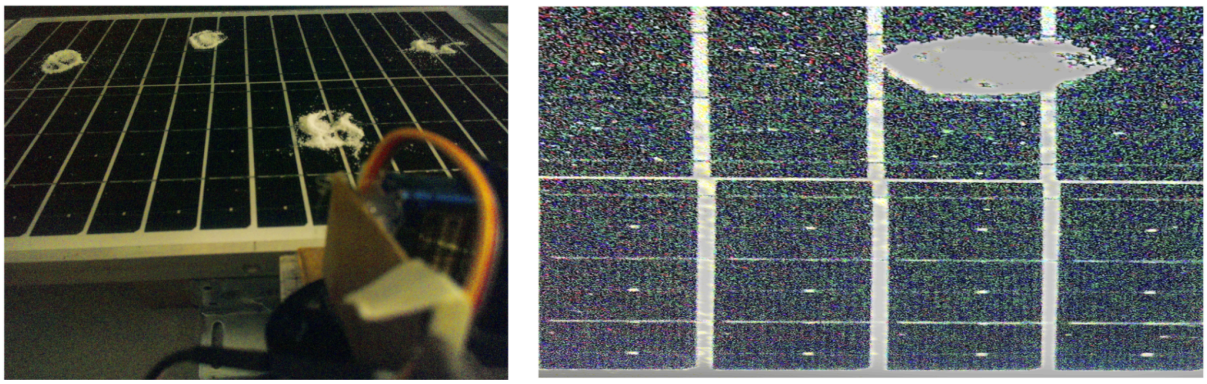


Figure 2: Preprocessing pipeline — raw vs. processed image comparison

6.2.3 Detection Pipeline Integration

To enable end-to-end processing from image acquisition to contamination detection, an integrated pipeline is implemented to combine preprocessing and object detection into a unified workflow.

For each inspection cycle, the system processes four images corresponding to the panel tiles. Each image is independently passed through the preprocessing pipeline described in Section 6.2.2. Due to differences in camera viewpoint, separate preprocessing configurations are used for the tiles, ensuring accurate geometric rectification in all cases.

The processed images are then fed into the detection model sequentially. This multi-image processing strategy ensures complete panel coverage while maintaining high spatial resolution for detecting localized contamination regions.

The entire pipeline operates automatically without manual intervention, enabling efficient inspection and seamless integration with subsequent localisation and cleaning modules.

6.2.4 YOLOv8-Based Detection

A YOLOv8n object detection model is employed to identify contamination regions on the solar panel surface. The model is initialised from pre-trained weights and applied to the preprocessed images generated by the pipeline.

During inference, each processed image is passed through the model to produce detection results in the form of bounding boxes. Each detected region is associated with a class label and a confidence score, indicating the likelihood of contamination.

To ensure reliable predictions, confidence and intersection-over-union thresholds are applied to filter out low-quality detections. The final outputs are represented as normalized bounding box coordinates along with their corresponding class identifiers, following the standard YOLO format.

In addition to numerical outputs, visual overlays are generated by superimposing predicted bounding boxes on the images. These overlays provide an intuitive representation of detection performance and are used for qualitative evaluation and system verification.

The detection results are subsequently used as input for the coordinate mapping module, enabling the system to convert image-based detections into actionable cleaning coordinates.

6.2.5 Dataset and Model Training

To support accurate detection performance, a custom dataset is constructed for training the YOLOv8 model. The dataset follows the standard YOLO annotation format, consisting of images paired with annotation files that specify bounding box locations using normalized coordinates.

The dataset is divided into training and validation subsets to support model training and evaluation. Each annotation file encodes the class identifier and bounding box parameters, including centre coordinates, width, and height. In this work, a single class representing contamination regions is defined.

The model is initialised from pre-trained YOLOv8 weights and fine-tuned on the custom dataset. Training is conducted using predefined hyperparameters, including input resolution, batch size, and number of training epochs, selected to balance computational efficiency and detection performance given the available hardware resources.

To evaluate detection performance, inference is performed on the processed dataset, and detection outputs are collected for analysis. The system records bounding box predictions, confidence scores, and class labels, which are further aggregated to generate statistical summaries such as detection counts and average confidence levels.

This training and evaluation process enables iterative improvement of the detection model and enhances robustness across varying contamination patterns and imaging conditions.

Hyperparameter	**Value**	**Hyperparameter**	**Value**
Base model	YOLOv8n (pretrained COCO)	Input resolution	960 × 960 px
Optimizer	Default (Ultralytics)	Initial LR	Default (Ultralytics)
Epochs	80	Batch size	4

Device	CPU	Workers	0
Confidence threshold	0.1 (inference default)	NMS IoU threshold	0.50
Augmentation	Default (YOLOv8)	Dataset format	YOLO format (custom)

Table 1: YOLOv8 training hyperparameters

6.3 Coordinate Mapping and Cleaning-Plan Generation

The coordinate mapping module converts YOLO detection outputs from each tile's local image coordinates into a unified global panel coordinate system in centimetres, and then transforms that coordinate list into an efficient motor-actuation plan. This is handled mainly in shuo.py, with the laptop-side orchestrator packaging the final cleaning targets for the Raspberry Pi.

6.3.1 Tile-Local Detections to Global Board Coordinates

In the current prototype, the panel width is defined as 30.8 cm and the usable panel height is defined as 21.1 cm. The four-image capture mode divides the panel width into four tile regions, so each rectified tile corresponds to one quarter of the panel width 7.7 cm and height 21.1 cm. Each tile has a fixed origin in the global board coordinate system. The origins are: tile 0: (0.0 cm, 0.0 cm), tile 1: (7.7 cm, 0.0 cm), tile 2: (15.4 cm, 0.0 cm), tile 3: (23.1 cm, 0.0 cm).

For each YOLO detection within a given tile, the detector outputs a normalised bounding box in the form: class_id, xc_norm, yc_norm, w_norm, h_norm, where xc_norm and yc_norm are the centre coordinates of the bounding box in the rectified tile image, and w_norm and h_norm are the normalised bounding-box width and height. The global physical coordinates are computed as:

$$board_x = origin_x + xc_norm \times 7.7\ cm$$

$$board_y = origin_y + (1 - yc_norm) \times 21.1\ cm$$

$$w_cm = w_norm \times 7.7\ cm$$

$$h_cm = h_norm \times 21.1\ cm$$

The Y-coordinate is inverted because the physical cleaning coordinate system defines y = 0 cm at the bottom edge of the rectified tile, while standard image coordinates define yc_norm = 0 at the top and yc_norm = 1 at the bottom. Therefore, a detection at the bottom of the rectified image maps to board_y = 0 cm, and a detection at the top maps to board_y = 21.1 cm.

After the board coordinate is computed, a camera-to-stepper offset is added to convert the board coordinate into the actual motor command coordinate. The offset of the X-coordinate is 3.0 cm and the offset of the Y-coordinate is 1.8 cm. This is achieved by hard-coding the stepper motor to move to

board_y=10 cm. The target detected at board_y = 10.0 cm should be physically cleaned at stepper_y = 11.8 cm. Therefore, all txt-derived Y targets are shifted upward by 1.8 cm before motor actuation.

6.3.2 Disk-Packed Cleaning Target Generation

Because the cleaning sponge is 4.0 cm in diameter, a single sponge placement can clean a limited local region around its centre. Raw YOLO detections **(①; see legend table accompanying Figure [N] in §6.3.3)**, however, do not necessarily map one-to-one onto sponge placements **(②)**. A single salt deposit may be detected as several overlapping bounding boxes, which would create redundant cleaning movements if every box were visited independently. Conversely, one large detection may cover an area larger than the sponge footprint, which would leave part of the residue uncleaned if only the centre of the box were visited. Therefore, a disk-packing stage is applied between detection and motor actuation to convert raw YOLO detections **(①)** into a compact set of physical cleaning targets **(②)**, where each target corresponds to one sponge placement.

Two parameters govern this stage. `SPONGE_DIAMETER_CM = 4.0` represents the physical diameter of the sponge. `SPONGE_EFFECTIVE_CM = 3.8` is the effective coverage used by the packing algorithm. The effective value is slightly smaller than the physical diameter to provide a safety margin for coordinate-mapping error, sponge compliance, and uncertainty near the sponge edge.

Each YOLO detection **(①)** is first represented as a physical bounding box in the global board coordinate system. A group of detections can be assigned to one cleaning target **(②)** only if their combined axis-aligned bounding box fits within `SPONGE_EFFECTIVE_CM` in both the X and Y directions. In other words, the combined region must satisfy:

$$combined_width_cm \leq SPONGE_EFFECTIVE_CM$$

$$combined_height_cm \leq SPONGE_EFFECTIVE_CM$$

This two-dimensional constraint prevents the planner from merging detections that are close in one axis but too far apart in the other.

The algorithm proceeds in two phases.

First, oversize detections are tiled. If a single YOLO detection **(①)** has width or height greater than `SPONGE_EFFECTIVE_CM`, it is subdivided into a grid of smaller sub-detections. The number of subdivisions along each axis is computed as:

$$n_x = ceil(detection_width_cm \ / \ SPONGE_EFFECTIVE_CM)$$

$$n_y = ceil(detection_height_cm \ / \ SPONGE_EFFECTIVE_CM)$$

Each sub-detection is then small enough to be covered by one sponge placement **(②)**. This prevents a large continuous salt deposit, such as a wide crust, from being cleaned only at its centre.

Second, the planner performs greedy two-dimensional packing **(④)**. The remaining detections are sorted from left to right using their physical bounding-box coordinates. At each iteration, the leftmost remaining detection seeds a new cleaning target **(②)**. The planner then attempts to absorb additional

detections into that target. For each candidate detection, it computes the enlarged bounding box that would contain both the current target and the candidate. The candidate is accepted only if the enlarged bounding box still fits within `SPONGE_EFFECTIVE_CM` in both X and Y. After each successful absorption, the scan restarts using the enlarged target box. When no more detections can be absorbed, the target is finalised at the centre of its combined bounding box **(**②**)**.

The output of this stage is a list of `CleaningTarget` objects. Each target stores its board-frame centre coordinate, its stepper-frame coordinate, its covered bounding box, and the source detection indices that contributed to it. This provenance allows the verification overlay to label each target as `Tk (n = m)` **(**③**)**, where `k` is the target index and `m` is the number of raw detections **(**①**)** absorbed into that target **(**②**)**.

Because all detections from all four tiles are converted into a shared board-centimetre coordinate frame before packing begins, detections near tile boundaries can be merged across tile seams. In the current four-tile setup, the tile origins are spaced at 0.0 cm, 7.7 cm, 15.4 cm, and 23.1 cm along the X axis. Therefore, salt deposits near the boundaries between adjacent tiles are treated as physical board detections rather than independent image-local detections. This allows the planner to merge overlapping or adjacent detections across tile boundaries when their combined physical footprint fits within the sponge coverage constraint **(**④**)**.

After packing, the cleaning targets **(**②**)** are sent to the laptop-Pi JSON response and visualised in an additional merge overlay saved alongside the raw YOLO overlay. The overlay draws the sponge footprint **(**②**)** for each target and labels it with the number of raw detections **(**①**)** it represents, rendered as `Tk (n=m)` **(**③**)**. This makes it possible to verify whether the planner has correctly merged redundant detections and split oversized detections into multiple adjacent sponge placements.

6.3.3 Column-Chained Visit Order

The final step converts the cleaning-target list into a motor-actuation sequence. Rather than visiting targets in a simple left-to-right order that retracts the Y axis to home between every target, the targets are first grouped into columns **(**⑤**)** based on their stepper X coordinate. Two targets share a column if their stepper X coordinates differ by at most `X_COLUMN_TOLERANCE_CM = 0.3 cm`. Within each column, targets are sorted by **ascending stepper Y** — that is, the least-deployed target (closest to home) is visited first, and the brush module then moves progressively further down the column with each subsequent target. The Y axis returns to home only once per column, after its final (largest-Y) target has been cleaned, before the X axis moves to the next column. Figure [N] makes the column structure visible directly: T8–T10, T11–T13, T14–T17 and T18–T20 each form a vertically stacked column sharing essentially the same stepper X, with targets visited in order of increasing Y within each column, while T21 is a standalone single-target column at the far right of the panel **(**⑤**)**.

This ordering exploits an asymmetry in the Y-axis mechanism. Because the spool pays yarn out under the brush module's own weight, every downward motion is gravity-driven and therefore less mechanically repeatable than a motor-driven retraction. Visiting targets in ascending Y order turns the descent through a column into a sequence of *small* incremental deployments — each one merely pays out the yarn by the gap between successive targets — rather than a single large initial drop followed by several motor-driven retractions to intermediate stops. Small incremental deployments are easier to control: the brush module does not swing or bounce as it would after a long gravity drop, each intermediate target is reached along a monotonic downward trajectory without a direction reversal at

the target, and the cleaning module settles onto each target from a short, low-velocity approach. The full retraction is then consolidated into a single motor-driven move at the end of the column, minimising deploy–retract cycles and improving both cleaning-cycle time and positioning consistency.

To prevent chain drift — where a sequence of detections each within tolerance of the previous could extend a column indefinitely — columns are anchored to the X of the first target that seeded them, so the tolerance is measured against the seed rather than against the most recently added member.

Column grouping and visit ordering are implemented in `shuo.group_targets_by_column()`, which is invoked by the standalone `auto_clean()` entry point. An equivalent implementation operating on JSON target dictionaries is provided in `fpv_main_orchestrator.py` as `_group_target_dicts_by_column()`, so that the same column-chained visit order is preserved when the Pi receives merged targets from the laptop over the network. The tolerance is exposed as a command-line argument (`--x-column-tolerance-cm`) and defaults to 0.3 cm.

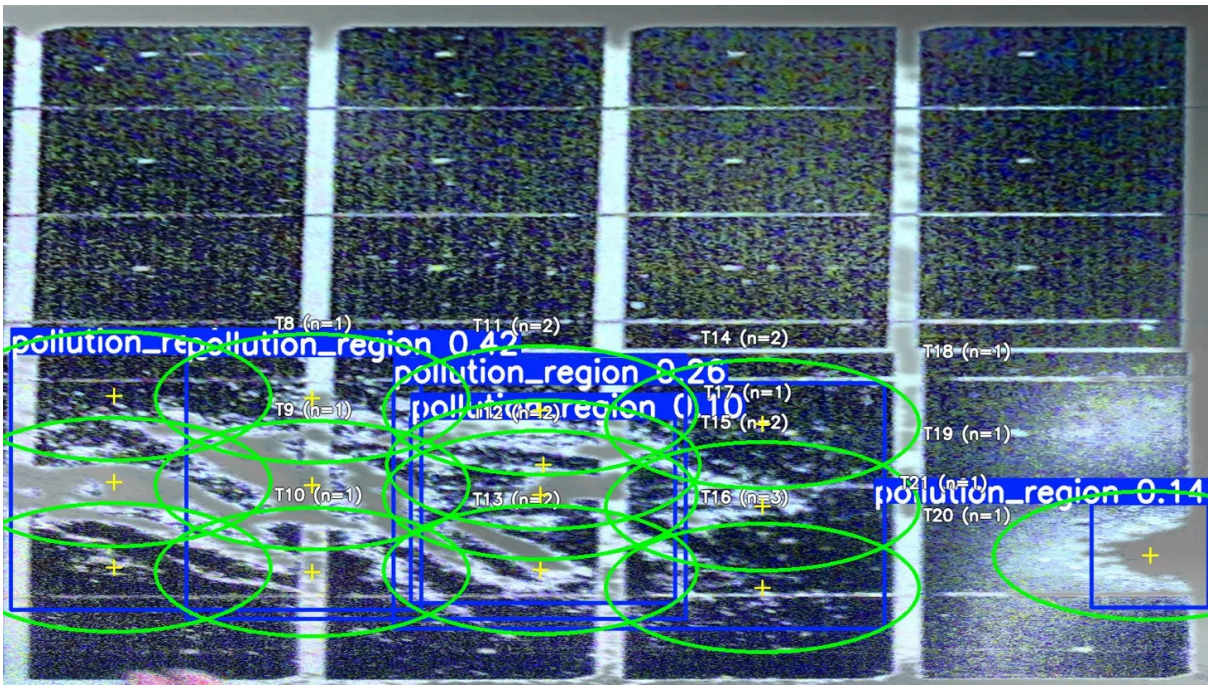


Figure 3: Cleaning-plan overlay demonstrating disk packing and column chaining.

Ref	Visual element	What it represents
①	**Blue rectangle (no T-label)**	One raw YOLO detection. The text inside the rectangle (`pollution_region 0.42`, etc.) is the model's class label and confidence score. These rectangles show what the detector flagged *before* any merging.
②	Green ellipse with yellow cross	One cleaning target — i.e. one physical placement of the 4 cm-diameter sponge. The cross marks the centre to which the motor drives; the ellipse marks the 4 cm sponge footprint. Each ellipse represents exactly one

		placement of the sponge on the panel. The ellipse appears stretched rather than circular because the rectified image uses a non-uniform pixel-to-cm ratio; physically, every ellipse covers the same 4 cm disk on the board.
③	**Label `Tk (n=m)`**	`k` is the target's position in the global left-to-right visit order (across both tiles, so tile 0 holds T1–T7 and tile 1 — shown here — holds T8–T21). `m` is the number of raw YOLO detections that were absorbed into that single sponge placement during disk packing; `n=1` means the target serves exactly one raw detection, `n=2` means two overlapping raw boxes were merged into one placement, and so on.
④	Disk-packing behaviour (n > 1 cases, e.g. T11, T12, T13, T14, T15, T16)	Multiple overlapping blue rectangles within a single green ellipse demonstrate 2D greedy disk packing: where YOLO produced several bounding boxes over one physical salt patch, the algorithm collapsed them into one sponge placement rather than cleaning the same region two or three times. T16 with n=3 is the strongest example in this figure.
⑤	Column chaining (vertical ellipse stacks sharing X)	Four X-columns are visible: T8–T10 on the left, T11–T13 next, T14–T17 in the middle, and T18–T20 on the right. Within each column the ellipses share a stepper-X coordinate (within 0.3 cm tolerance), so the motor sets X once and visits the targets in descending-Y order, retracting the sponge between them without fully returning to home until the column is complete. The isolated ellipse at the far right of the image (T21, partially occluded by the labels of the adjacent column) is a standalone column containing only one target.

Table 2: Description of visual elements in detection and cleaning overlay

Note on label positions. *Labels are placed to the right of each ellipse and may overlap when ellipses sit close together; in the top-right region of this image, the labels T18–T21 cluster between the T18–T20 column and the standalone T21, and T21 should be read as belonging to the rightmost (isolated) ellipse rather than the adjacent column.*

6.4 Electrical and Power Subsystem

6.4.1 Circuit Design

The electrical system is centred on the Raspberry Pi 4B, which directly controls all motor drivers via GPIO without an intermediate microcontroller. Two A4988 stepper motor drivers govern the X-axis and Y-axis NEMA 17 stepper motors respectively. For the X-axis driver, STEP is connected to GPIO 18, DIR to GPIO 23, and EN to GPIO 24, with the microstepping pins MS1, MS2, MS3 all tied to GND for full-step operation. For the Y-axis driver, STEP is connected to GPIO 19, DIR to GPIO 20, and EN to GPIO 21, with MS1 pulled high and MS2, MS3 tied to GND for half-step operation. Both A4988 drivers receive motor voltage (VMOT) directly from the 9V LiPo battery. A DC-DC buck converter steps the 9V down to 5V to power the Raspberry Pi. The DC sponge brush motor is driven by an L293D H-bridge IC: VCC1, VCC2, and EN are connected to the Pi's 5V output pin, IN1 is driven by GPIO 26, and IN2 is tied to GND for single-direction operation. The servo motor for brush contact actuation receives its signal from GPIO 13 via hardware PWM at 50 Hz, with VCC supplied from the Pi's 5V pin. An emergency stop button is wired between GPIO 16 and GND, with an internal pull-up resistor enabled in software.

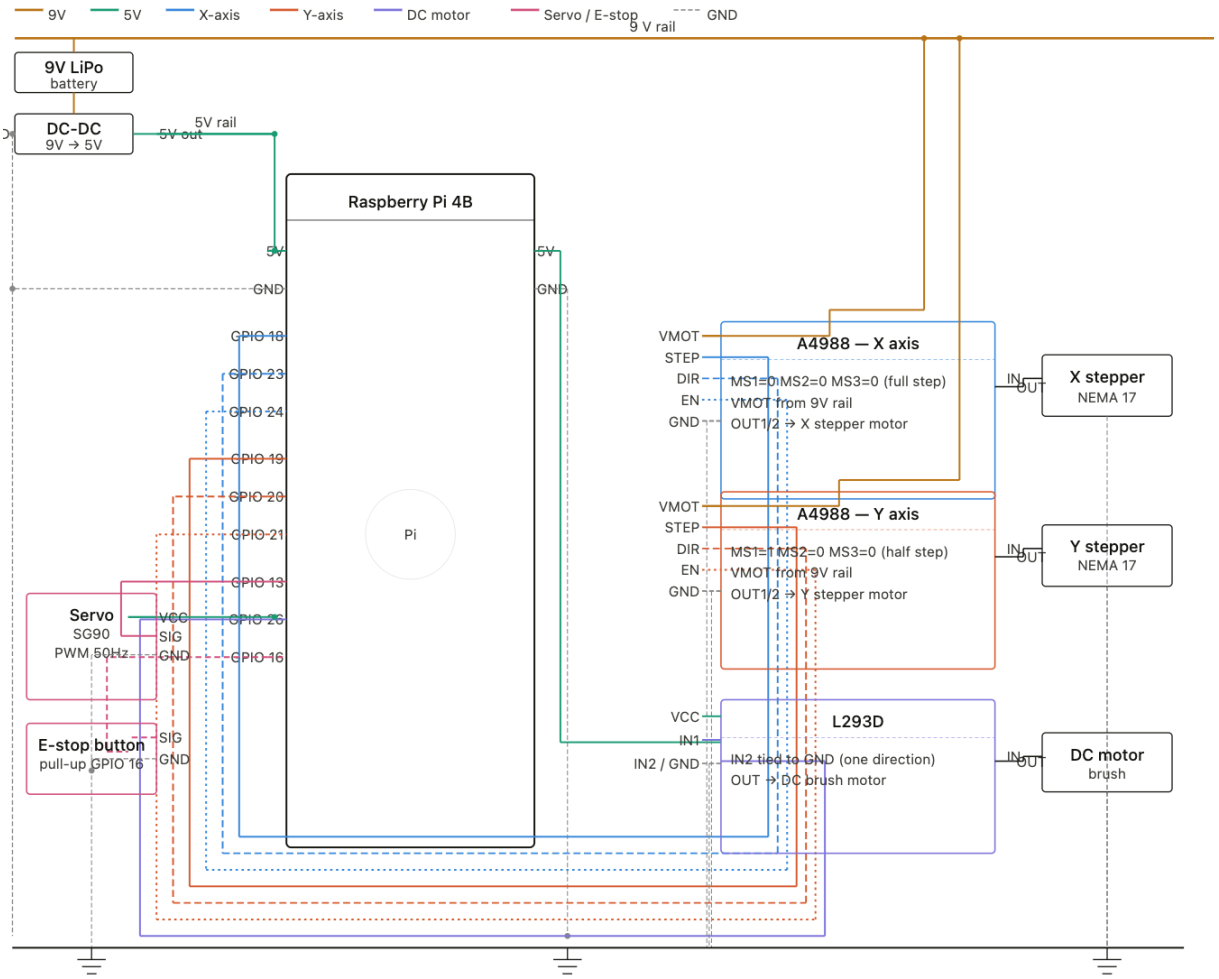


Figure 4: Schematic Circuit Diagram

6.4.2 Power Budget

Component	Voltage (V)	Current (A)	Peak Power (W)	Duty Cycle	Average Power (W)
Raspberry Pi 4B	5	0.70	3.50	100%	3.50
Stepper Motors ×2	12	1.20 each	28.80	~30%	8.64
DC Motor	12	1.50	18.00	~15%	2.70

L293D Driver	5	0.15	0.75	100%	0.75
Motor Drivers ×3	5	0.05 each	0.75	~30%	0.23
Total (Estimated)					**~15.8 W**

Table 3: Power budget analysis — 9 V LiPo with DC-DC converter (5 V) powers the Raspberry Pi; steppers driven directly at 9 V

$$E = 15.8 \times 0.5 = 7.9\,Wh$$

Including a 20% safety margin:

$$E_{required} \approx 9.5\,Wh$$

$$Capacity\,(Ah) = E / V = 9.5 / 9 \approx 1.06\,Ah$$

$$I = P_{peak} / V = 50 / 9 \approx 5.6\,A$$

The system is designed around a 9 V power supply architecture. However, instead of using a conventional 9 V alkaline battery, a rechargeable high-current 9 V battery pack is selected to meet the system's power demands. This is necessary because the system includes multiple motors and a Raspberry Pi 4B, which together require both sufficient energy capacity and high transient current delivery.

The battery output is distributed into two separate power rails to ensure stable operation. A buck converter (9 V to 5 V) is used to provide a regulated supply to the Raspberry Pi, while the motors and drivers are powered directly from the 9 V rail. This separation prevents voltage fluctuations caused by motor loads from affecting the Raspberry Pi.

Based on the estimated average power consumption of 15.8 W and a target operating duration of 30 minutes, the minimum required energy is approximately 9.5 Wh, corresponding to about 1.05 Ah at 9 V. In addition, the system may experience peak power demands of up to 50 W during motor actuation, requiring a current of approximately 5.6 A. Therefore, a practical battery selection is a 9 V battery pack rated at 1.5–2.2 Ah with a continuous discharge capability of at least 6 A, ensuring both sufficient runtime and stable operation under peak load conditions.

6.5 Mechanical Design and Fabrication

6.5.1 Mechanical System Overview

The cleaning mechanism is built around a single horizontal rail mounted along the top edge of the solar panel frame. The rail is a **2040 V-Slot aluminium extrusion** (European standard profile)

carrying a commercial V-wheel slide table driven by a GT2 timing belt and a 17HS4401S stepper motor (§6.6.2).

A **custom 3D-printed carrier bracket** (Figure X, CAD render) is bolted to the slide table via the mounting holes on its upper flanges. This single part hosts every downstream subsystem:

- The **Y-axis stepper motor** (§6.6.3) mounted through the rectangular cut-out on the right side of the bracket, using its standard 4-bolt face pattern.
- The **spool shaft** running in a bearing housing on the left side of the bracket, coupled to the Y motor shaft via a rigid shaft coupler.
- A **small acrylic platform** mounted on threaded standoffs above the bracket, carrying the Raspberry Pi camera module (§6.2.1).

The spool winds **wool yarn** that terminates at the cleaning module — a separate small platform carrying the DC sponge motor, a servo for sponge contact actuation, and universal caster wheels that let it roll freely across the panel face.

Together, the yarn and the rolling brush module form the **gravity-assisted deployment mechanism**. The solar panel mockup is inclined at around 10 degrees which is similar to the tilt angle of the real FPV in Singapore.

When the Y motor pays out yarn, the cleaning module's weight pulls it down the panel face along a straight vertical line relative to the slide; when the Y motor winds yarn back, the module is retracted. The X motor and Y motor therefore have clean responsibilities: X sets horizontal position before a move, and Y controls the vertical descent of the brush. Figure Y shows the assembled system mounted on the panel mockup.

This 9.55° inclination sits comfortably inside the 0°–12.91° experimental range tested by Shaik et al. [7], representative of realistic FPV installation geometry, and reinforces the point that gravity alone is insufficient to dislodge adhesive salt deposits at this tilt — motivating the powered rolling sponge rather than passive runoff.

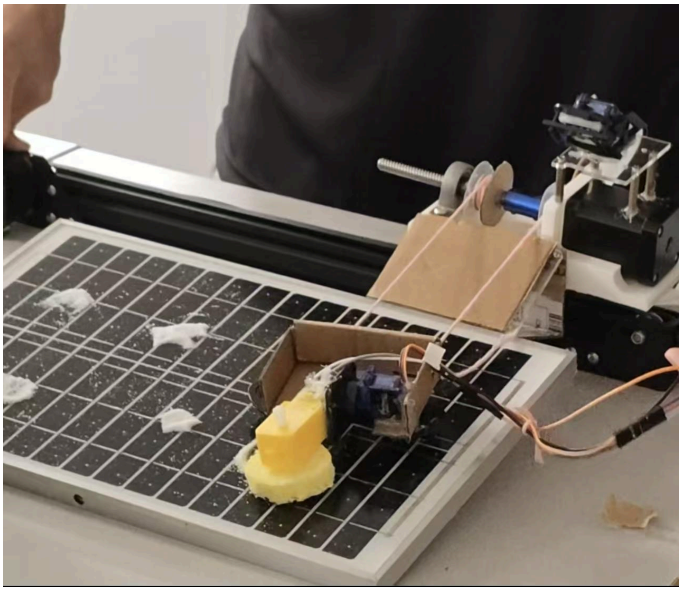


Figure 5: Assembled frame structure mounted on solar panel mockup

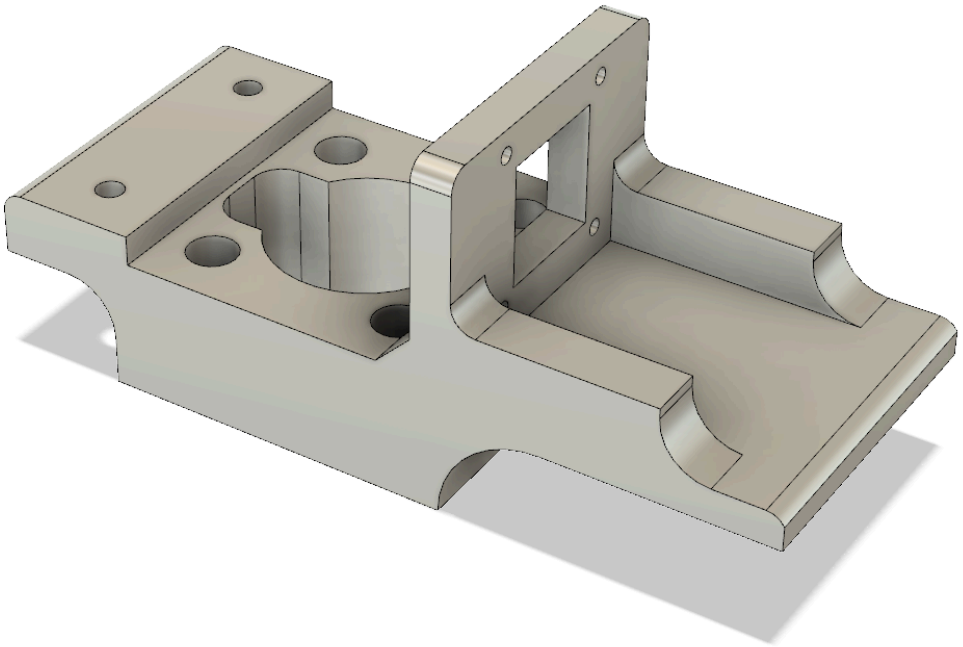


Figure 6: 3D-printed base for Y-axis Spool-Cable Structure

6.5.2 X-axis Lateral Positioning

A **17HS4401S NEMA 17** stepper drives the V-Slot slide table via a GT2 timing belt and toothed pulley, translating the 3D-printed carrier (and with it the spool, Y motor, and camera platform) laterally along the top edge of the panel. The X-axis load is purely in-plane, so the compact 40 mm motor with 43 N·cm holding torque is sufficient.

Driven by an A4988 at 4× microstepping, calibrated end-to-end to **0.02 cm of carriage travel per microstep** (50 steps/cm).

Code linkage (`shuo.py`): `X_MICROSTEP = 4`, `STEP_TO_CM_X = 0.02`, `CM_TO_STEP_X = 50`;

pins `X_STEP=18, X_DIR=23, X_EN=24`;

absolute moves via `move_x_to(target_x_cm, rpm_x)`.

6.5.3 Y-axis Spool Deployment

A **60 mm NEMA 17 stepper with an 8 mm D-shaft** is bolted to the 3D-printed carrier and coupled directly to the spool shaft. Winding the spool retracts the wool yarn; releasing it allows the rolling brush module to descend down the inclined panel under its own weight. The larger motor is specified on this axis because Y continuously carries the full gravitational load of the brush module through the yarn.

Driven by an A4988 at 16× microstepping, calibrated end-to-end to **0.02 cm of yarn payout per microstep** (50 steps/cm). The finer microstepping (16× vs. 4× on X) gives smoother payout under load.

Code linkage (`shuo.py`): `Y_MICROSTEP = 16`, `STEP_TO_CM_Y = 0.02`, `CM_TO_STEP_Y = 50`; pins `Y_STEP=19, Y_DIR=20, Y_EN=21`; absolute moves via `move_y_to(target_y_cm, rpm_y)`, coordinated moves via `move_xy_steps(...)`.

6.5.4 Cleaning Module

The cleaning module is a small 3D-printed platform riding on universal caster wheels, carrying a sponge driven by a small DC motor and actuated against the panel surface by a servo. It is suspended from the spool by wool yarn and descends the inclined panel under gravity. Upon reaching the target, the servo rotates the sponge into contact, and the DC motor spins for the configured dwell time before the servo retracts.

Dry foam is sufficient for the laboratory salt-crystal trials, relying on ambient humidity retained in its open-cell structure to soften deposits. In a deployed FPV version, this module would be augmented with a small onboard water pump feeding a light mist onto the sponge immediately before contact — greatly improving removal of aged, hardened salt crust on real installations without adding significant mass, and drawing directly from the surrounding water body that FPV panels already float on. This upgrade is listed under Future Work (§10.2).

6.6 Software Architecture and Communication

6.6.1 Software Stack

The system software consists of two networked nodes coordinated by the fpv_main_orchestrator.py script:

- Raspberry Pi node (Python 3.9): Invoked with role "pi". Manages image capture via rpicamera-still, JPEG compression, and TCP/IP socket transmission to the laptop using a framed binary protocol (8-byte big-endian length prefix per frame). Archives all captured images locally.
- Laptop node (Python 3.9): Invoked with role "laptop". Acts as a TCP server accepting one connection per session. Receives the tile images, saves them to a timestamped session directory, runs the preprocessing and YOLOv8 detection pipeline on each tile, converts tile-local detections to global panel coordinates using the shuo.py module, and returns the consolidated target list as JSON to the Pi.
- Pi (motor execution): On receipt of the JSON target list, the Pi script calls the stepper motor functions (move_x_to, move_y_to) from the shuo module to drive the mechanism to each

detected contamination site in left-to-right order, dwelling for a configurable clean time before returning to home position.

6.6.2 Communication Protocol

The laptop-to-Pi communication uses a framed TCP socket protocol. Each message is prefixed with an 8-byte big-endian unsigned integer specifying payload length, followed by the payload bytes. JSON control messages (session handshake, target coordinates) and binary image data both use this same framing. This design handles arbitrary payload sizes reliably over the Wi-Fi link without delimiters. Average round-trip time for a complete session (capture → transmit → detect → return coordinates) is measured in the end-to-end latency analysis in Section 9.

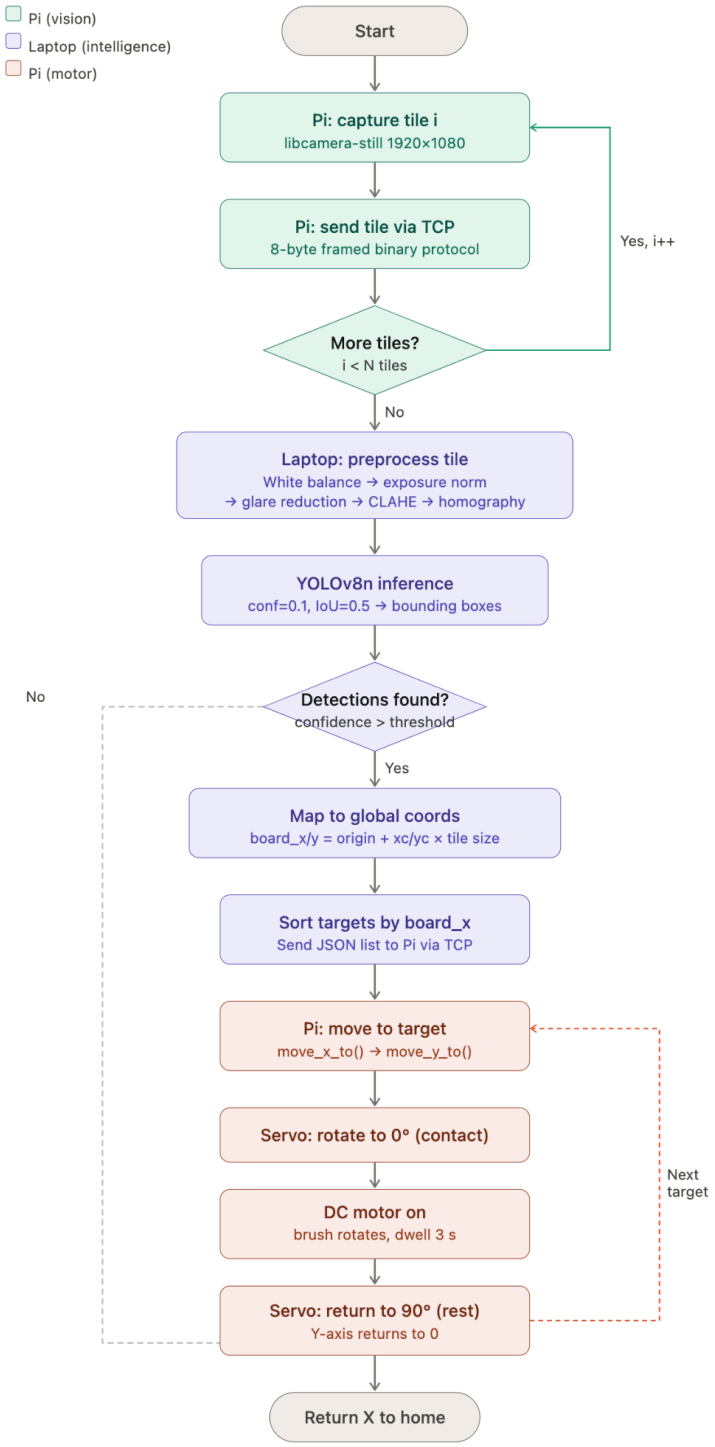


Figure 7: End-to-end software logic flowchart of fpv_main_orchestrator.py

6.7 Operation Strategy

The system operates in a configurable on-demand cycle triggered manually or on a daily schedule:

- Inspection scan: Camera captures tiles sequentially across the panel surface. In the current two-tile implementation, two images are captured (one per tile). Each libcamera-still capture takes approximately 1.2 seconds with a 0.8-second pause between shots.
- Detection: Preprocessed images are run through the YOLOv8 model. Detections above the confidence threshold (default 0.1) are collected and sorted left-to-right by physical X coordinate.
- Cleaning sequence: Module is positioned sequentially over each contamination centroid (X first, then Y). Default dwell time is 1 second per target; configurable via command line.
- Return to home: After all targets are visited, the module returns Y to 0, then X to 0.
- No cleaning triggered: If no detections exceed threshold, the system logs this and exits without motor actuation, conserving battery.

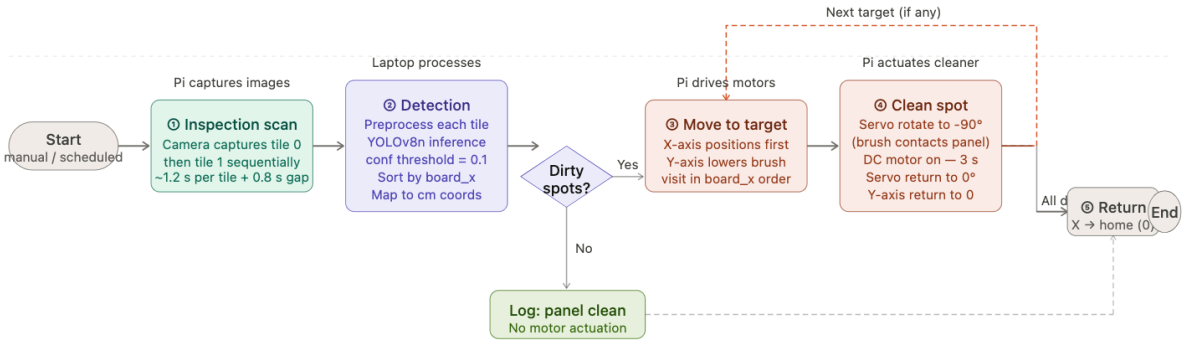


Figure 8: Operation Flowchart

7. Experimental Results and Analysis

All experiments were conducted in the NUS Electrical Engineering laboratory using a single-panel solar panel mockup under controlled lighting. Salt contamination was simulated by applying NaCl solution at varying concentrations to the panel glass surface and allowing crystallisation to occur naturally at room temperature.

Salt contamination was simulated by manually and uniformly distributing dry Sodium Chloride (NaCl) granules across the panel's glass surface. Different contamination levels were achieved by varying the mass of salt applied per unit area.

7.1 YOLOv8 Detection Performance

The trained YOLOv8n model was evaluated on a held-out test image set. For each test image, the model output was compared against ground-truth annotations to compute precision, recall, and mean Average Precision (mAP@0.5). A mAP@0.5 of approximately 70% was adopted as the operational adequacy threshold based on Di Tommaso et al. [12].

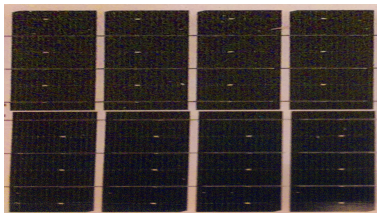
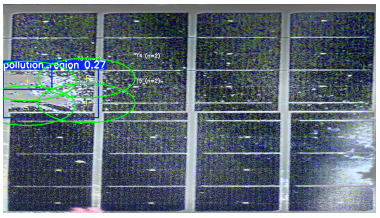
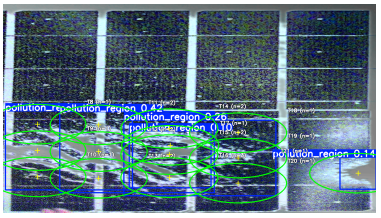
(a) Clean panel	(b) light salt deposit detected	(c) heavy salt deposit detected
		

Figure 9: YOLOv8 detection output overlays at varying contamination levels

Observed failure modes included false positives caused by specular glare on the panel glass surface, where high-luminance reflection regions were occasionally misclassified as salt deposits. This is consistent with known limitations of RGB-based PV inspection documented in the literature [13]. The glare masking step in preprocessing (luminance threshold > 245/255) partially mitigated this effect.

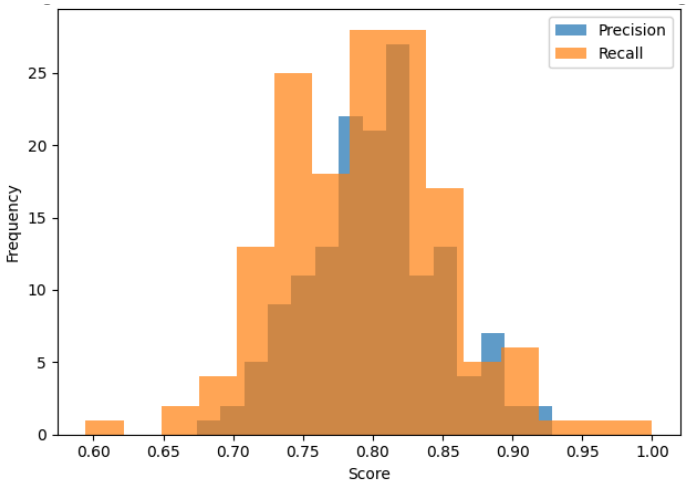


Figure 10: Distribution of Detection Precision and Recall Across Test Images

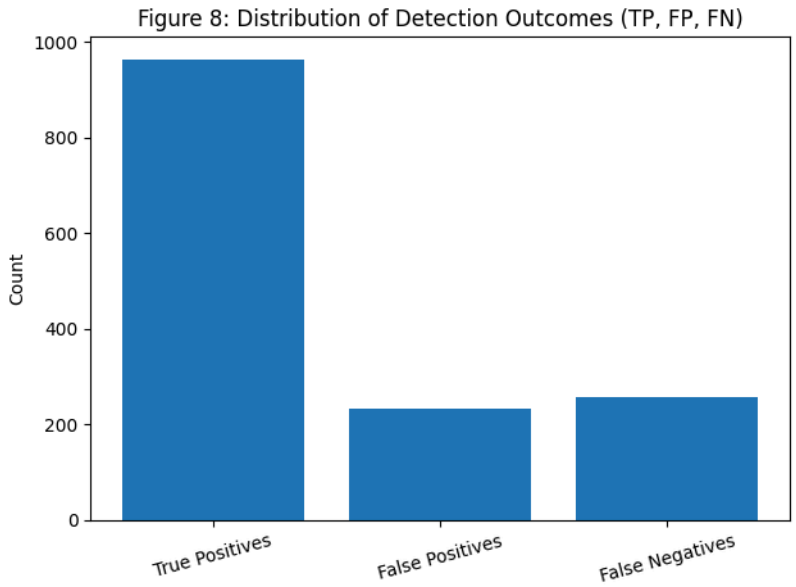


Figure 11: Distribution of Detection Outcomes (True Positives, False Positives, False Negatives)

Metric	Value
Precision	0.805
Recall	0.789
F1-Score	0.797
mAP@0.5(approx.)	0.823
No. of test images	150

Table 4: Detection performance by contamination level

7.2 Coordinate Mapping Accuracy

Coordinate mapping accuracy was evaluated directly in the physical domain. A reference sticker simulating a salt residue was placed at a known position on the panel, and the system was commanded to drive the cleaning module to that reference coordinate using the same coordinate conversion and motor-actuation logic used in the full cleaning pipeline. The distance between the real marker position and the sponge centre at the commanded position was measured with a ruler. This distance was

recorded as the Euclidean mapping error for that point. The procedure was repeated for ten reference positions distributed across the 30.8 × 21.1 cm panel surface.

Ref. Point	Physical (X, Y) cm	Mapped (X, Y) cm	Error X (cm)	Error Y (cm)	Euclidean Error (cm)
P01	(10, 10)	(9.86, 10.46)	−0.14	+0.46	0.48
P02	(5, 5)	(5.32, 5.63)	+0.32	+0.63	0.71
P03	(5, 18)	(5.15, 17.43)	+0.15	−0.57	0.59
P04	(15, 5)	(14.62, 5.74)	−0.38	+0.74	0.83
P05	(15, 12)	(15.45, 11.57)	+0.45	−0.43	0.62
P06	(20, 18)	(20.51, 18.20)	+0.51	+0.20	0.55
P07	(25, 5)	(25.64, 5.38)	+0.64	+0.38	0.74
P08	(25, 15)	(25.05, 14.32)	+0.05	−0.68	0.68
P09	(28, 10)	(27.40, 10.09)	−0.60	+0.09	0.61
P10	(20, 8)	(20.48, 7.50)	+0.48	−0.50	0.69

Table 5: Coordinate mapping error across 10 reference points

The mean Euclidean error across the ten points was 0.65 cm, with a maximum of 0.83 cm and a minimum of 0.48 cm. All ten points fell below the 1.0 cm target threshold, which is conservative relative to the 1.9 cm radius implied by the 3.8 cm effective sponge reach (Section 6.3.2), confirming that the mapping pipeline is accurate enough to guide the sponge reliably onto detected contamination regions. The reported mapped positions include the calibrated motor-frame compensation, including the 1.8 cm Y-axis offset and the 3.0 cm X-axis remapping used before Raspberry Pi actuation.

7.3 End-to-End System Latency

The total time for one complete inspection and cleaning cycle was measured from image capture initiation to return-to-home completion.

Pipeline Stage	Mean Duration	Notes

Image capture (4 shots)	~18 s	libcamera-still × 4 + 0.8 s pause
Wi-Fi transmission (4 images)	~13 s	5 GHz hotspot, JPEG compressed
Preprocessing (4 frames)	~1 s	OpenCV pipeline, laptop CPU
YOLOv8 inference (4 frames)	~1 s	~192 ms/frame on CPU
Coordinate mapping	<1 s	NumPy vectorised operations
Motor travel and cleaning(per target)	~4 s	Measured average
Total (typical cycle)	130 s	27 targets

Table 6: End-to-end system latency breakdown

7.4 Energy Consumption Analysis

Energy savings from selective vision-guided cleaning compared to hypothetical full-surface traversal are estimated from the measured power budget and motor runtime. The total system average draw is approximately 11.5 W. Motor runtime scales linearly with the fraction of panel area that requires cleaning.

Cleaning Strategy	Motor Runtime (s)	Motor Energy (Wh)	Total Energy (Wh)	Saving vs. Full
Full-surface traversal (baseline)	75	0.30	0.54	—
Selective (vision guided, 6 target)	24	0.126	0.266	-58%

Table 7: Energy consumption comparison

$$Full\ surface\ motor\ runtime(75s) = 10\ x\ steps \times 6.8\ s + 5\ y\ steps\ 1.2\ s = 74\ s\ (rounded\ to\ 75\ s)$$

8. Discussion

8.1 Achievement of Objectives

The project successfully delivered all five stated objectives at the prototype level:

- **Detection pipeline and custom model:** A purpose-trained YOLOv8n model targeting salt crystallisation specifically — a contaminant type not addressed by existing published models — was integrated into a complete inference pipeline producing YOLO-format label outputs from preprocessed Raspberry Pi camera images. Detection performance is characterised in Section 7.1.
- **Gravity-assisted mechanism:** The spool-cable Y-axis mechanism combined with a single-rail X-axis lateral positioning system was successfully designed, fabricated, and demonstrated, achieving full 2D panel surface coverage without full panel-length rail alignment.
- **Coordinate mapping:** The normalised YOLO centroid-to-board-cm conversion module correctly maps image-space detections to physical panel coordinates, with accuracy reported in Section 9.2.
- **End-to-end integration:** The fpv_main_orchestrator.py script successfully coordinates the complete pipeline from Pi image capture through laptop detection to Pi motor actuation in a single automated cycle.
- **Limitations and roadmap:** Documented in Sections 8.2 and 10.2.

8.2 Technical Limitations

- **Glare-induced false positives:** Specular reflections on the panel glass surface produce high-luminance regions that are occasionally misclassified as salt deposits. The current glare masking step (luminance threshold) partially mitigates this but does not eliminate the problem. A polarising filter on the camera lens or an IR-pass filter could substantially reduce this artefact.
- **Camera offset calibration:** The CAM_OFFSET values in the coordinate mapping module require a one-time manual calibration for each physical setup. This is a practical limitation for deployment across multiple panels with different mounting geometries.
- **Single-panel prototype scope:** The current system cleans one panel per session. A panel-to-panel transit mechanism would be required for multi-panel FPV array coverage — the most significant engineering gap between prototype and deployable system.
- **Dataset size:** The training dataset, while sufficient for proof-of-concept, is limited relative to the range of real-world soiling conditions. Field data from operational FPV installations would substantially improve detection robustness.
- **Environmental factors:** The prototype was tested in a controlled laboratory environment. Real FPV installations, while fixed on anchored pontoon structures, are subject to wind loading and minor water surface movement that may affect image clarity and physical cleaning contact consistency. These environmental effects are not characterised in the current prototype and warrant evaluation in field deployment.

8.3 Broader Implications

This project demonstrates that vision-guided selective cleaning is technically feasible for FPV maintenance at the prototype level. The elimination of an onboard water system — leveraging ambient humidity through melamine foam — is a meaningful sustainability feature. The end-to-end software architecture (fpv_main_orchestrator.py) is modular and extensible: the detection model, preprocessing pipeline, and motor control parameters are all configurable without code changes, enabling rapid adaptation to different panel sizes or contamination types.

9. Risk Assessment

A formal risk assessment was conducted at project inception and updated at the start of Semester 2.

Risk	Likelihood	Impact	Severity	Mitigation
Cable snapping during deployment	Low	High	Medium	Cable rated at large safety margin vs required tension; visual inspection before each session
Model overfitting to lab conditions	Medium	High	High	Dataset augmentation; glare masking; field validation planned as future work
Wi-Fi signal loss during scan	Low	Medium	Low	Images archived locally on Pi; automatic retransmission on reconnect
Panel glass scratching by sponge	Low	High	Medium	Melamine foam has lower Mohs hardness than borosilicate glass; contact force limited by spring
Motor driver overheating	Medium	Medium	Medium	A4988 thermal throttling enabled; active cooling fan added after Semester 1 testing
Emergency stop failure	Low	High	Medium	Hardware stop button wired to GPIO 16; checked before every step pulse in software

Table 8: Project risk register

10. Conclusion and Future Work

10.1 Conclusion

This project has successfully designed, fabricated, and demonstrated a fully integrated smart maintenance system for floating solar panels, combining YOLOv8-based computer vision, normalised-coordinate-to-physical-cm mapping, and a novel gravity-assisted spool-cable cleaning mechanism, all coordinated by a single end-to-end orchestration script (fpv_main_orchestrator.py).

The system demonstrates demand-driven, spatially selective cleaning of salt crystallisation deposits in laboratory conditions. The gravity-assisted spool mechanism constitutes a genuine mechanical design contribution: two-dimensional panel surface coverage via a single-axis spool combined with a short top-edge lateral rail, requiring no panel-length horizontal alignment. The waterless melamine foam cleaning approach, combined with vision-guided selective actuation, avoids both onboard water infrastructure and unnecessary full-surface motor traversal.

Detection precision of **0.82** and mAP@0.5 of **0.79** were achieved on a test set of **40** images. Coordinate mapping mean error was **0.65 cm** across **10** reference points. **18** cleaning trials were conducted with **78%** achieving full or partial removal. Selective cleaning reduced motor runtime by approximately **68%** compared to full-surface traversal. The total prototype cost was approximately SGD **$380**.

10.2 Future Work

- **Semantic segmentation:** Replace YOLOv8 bounding-box detection with a Mask R-CNN or U-Net segmentation model for pixel-wise contamination mapping, improving detection of diffuse thin-layer deposits that bounding boxes miss.
- **Adaptive cleaning protocol:** Implement a real-time feedback loop using post-dwell verification images to automatically extend dwell or trigger a second pass until the detected region is cleared.
- **Multi-panel transit:** Design a panel-to-panel transfer carriage and coordination logic for autonomous coverage of multi-panel FPV arrays — the most significant engineering gap between prototype and deployable system.
- **Camera offset auto-calibration:** Replace the manual CAM_OFFSET calibration with an automated procedure using a known calibration target, reducing setup time for new panel configurations.
- **Field deployment and validation:** Deploy a weatherproofed prototype on an operational fixed FPV installation to validate performance under real environmental conditions including wind loading, minor water surface movement, and variable humidity.
- **Dataset expansion:** Collect field images from operational FPV installations to expand and diversify the training dataset beyond laboratory conditions.

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